PRE-CLASS READING GUIDE

Python Libraries, Modules & Packages • Working With Files

For Data Science / Data Engineering and Cybersecurity students

PURPOSE

Before the next practical class, study the two topics below. Both groups need these foundations to understand how different Python files work together and how programs read or save data. This is preparation material, not a project submission.

PART 1 | PYTHON LIBRARIES, MODULES & PACKAGES

Start here: A module is usually a Python file that can be imported. A package organizes related modules. Python includes a standard library, while third-party packages are installed separately. Your team will also write its own custom modules.

READ FIRST Primary reading: Python Modules and Packages — Real Python

A guided introduction to organizing programs into multiple Python files and importing what you need.

Focus on: What a module is; import and from ... import; package folders; the purpose of __init__.py; why modular design helps.

REFERENCE Reference: The Python Tutorial — Modules

The official Python explanation of modules and packages. Use it to revisit terminology and examples.

Focus on: Creating and importing your own .py module; packages and package imports.

OPTIONAL Extra help: Installing Packages — Python Packaging User Guide

Explains pip and how separately distributed Python packages are installed.

Focus on: The difference between importable modules and installed distributions; when and how pip is used.

After Part 1, you should be able to answer

- What is the difference between a library, module, and package?
- What is the difference between a Python standard-library module and a third-party package?
- Why does a project have a main.py file and other .py files?
- What does import do, and why might a package include __init__.py?

PART 2 | WORKING WITH FILES IN PYTHON

Start here: Programs often need to read saved data or write results. Files can contain plain text, CSV tables, and other formats. You will first learn basic text-file operations, then apply the ideas to your own project.

READ FIRST Primary reading: Reading and Writing Files in Python — Real Python

Introduces file objects, opening and closing files, file modes, reading/writing, and paths.

Focus on: open(); read(); write(); with open(...); r, w, a; relative file paths.

REFERENCE Reference: Python Tutorial — Reading and Writing Files

The official Python documentation explains file modes and why using with is good practice.

Focus on: Difference between reading, replacing and appending; automatic closing with with; text files versus binary files.

After Part 2, you should be able to answer

- What do the file modes r, w, and a mean? Which mode can overwrite existing content?
- Why is with open(...) useful?
- What is the difference between a relative file path and an absolute path?
- How can you verify that your program successfully created or updated a file?

HOW TO STUDY BEFORE CLASS

- Read the two READ FIRST resources in order; use the REFERENCE links to clarify anything confusing.
- Do not just skim. Open your Python editor and type at least one simple example from each topic.
- Try creating two small .py files and importing something from one into the other.
- Try creating a text file, writing a short message, then reopening it to read the message.
- Bring one question from each topic to the group session.

QUICK PREPARATION CHECKLIST

☐ I can explain a module and a package in simple words.

☐ I can distinguish built-in/standard-library and third-party tools.

☐ I understand what import does.

☐ I can explain reading, writing and appending to a file.

☐ I tried a short exercise on my own computer.

IMPORTANT FOR YOUR PROJECT

These readings are shared foundations for both departments. For the actual group assignment, follow only the project-specific libraries and functions listed in your department's project workbook. You do not need to learn unrelated cybersecurity or data-science libraries tonight.